

Task 1:

Main Components Inside a Desktop PC

Here is the list of the primary components found inside a desktop PC cabinet, along with a brief description of their functions:

- **1. Motherboard:** The main printed circuit board that connects all hardware components together, allowing them to communicate and function as a single system.
- **2. CPU (Central Processing Unit):** Often called the brain of the computer, it executes logic, performs calculations, and manages data flow across the system.
- **3. RAM (Random Access Memory):** The temporary high-speed workspace of the PC that stores data for currently running applications to ensure smooth

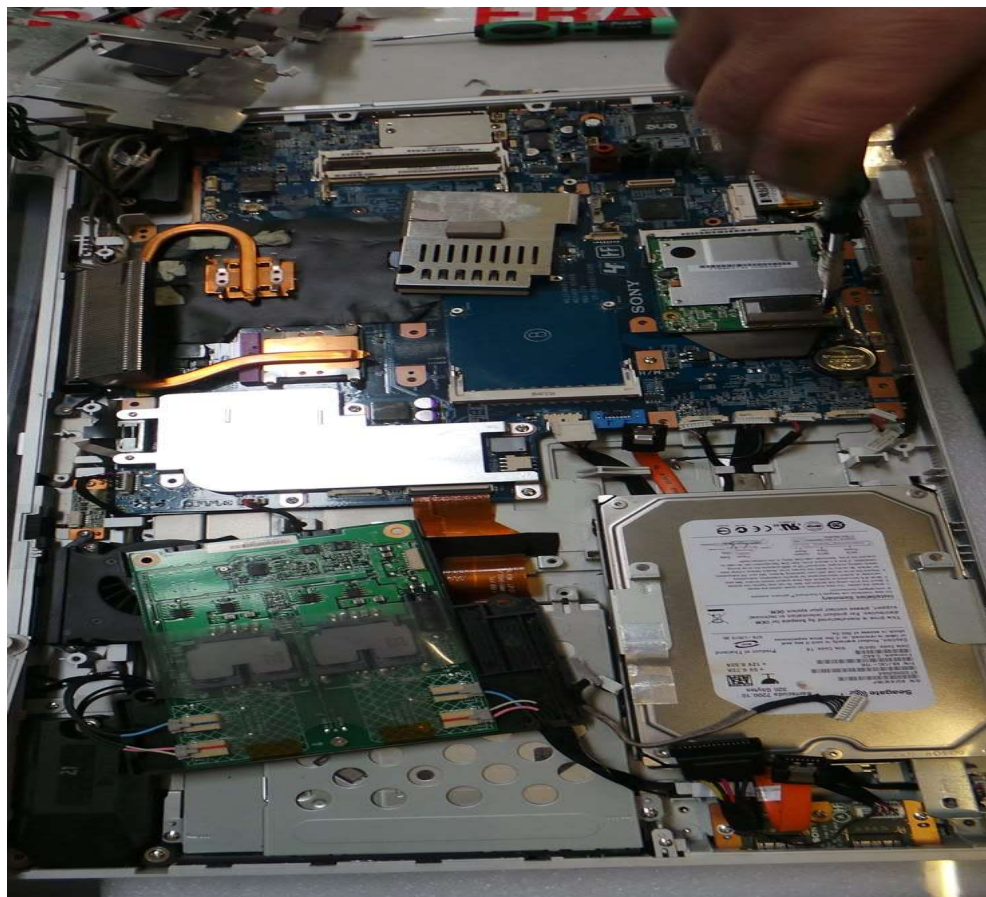
multitasking.

- **4. Storage Drive (SSD/HDD):** The permanent storage component where the operating system, software applications, and user files are saved.
- **5. Power Supply Unit (SMPS):** Converts main AC power from the wall outlet into stable DC power and distributes it to all internal hardware components.
- **6. GPU (Graphics Processing Unit):** Handles the rendering of images, videos, and heavy graphics; it can be integrated into the CPU or installed as a dedicated card.
- **7. Cooling System (Fans & Heatsink):** Essential thermal components that absorb and blow away the heat generated by the processor to prevent system overheating.

Task 2:

PC Disassembly Process (Step-by-Step Documented)





Below is the step-by-step documented process of disassembling a desktop PC safely, along with explanations for each action taken:

Step 1: Power Isolation and External Disconnection

- *(Insert a photo of unplugging cables from the back of a PC here)*
- **Action:** I turned off the SMPS switch and disconnected the main AC power cord, followed by all external peripherals (mouse, keyboard, monitor).
- **Reason:** This is the most critical step to ensure complete electrical safety and avoid any static shocks before touching internal parts.

Step 2: Removing the Side Panel

- *(Insert a photo of an open PC cabinet here)*
- **Action:** I unscrewed the two thumb screws at the back of the cabinet and carefully slid the left side panel off to expose the internal hardware.

Step 3: Disconnecting Internal Cables

- **Action:** I carefully unplugged the thick ATX 24-pin motherboard power cable, the 8-pin CPU power cable, and the SATA data cables connected to the storage drives.
- **Reason:** All connections must be safely removed before any hardware is pulled out to prevent cable snapping or port damage.

Step 4: Removing the Hardware Components (RAM, GPU, and Storage)

- *(Insert a photo of removing RAM or GPU here)*
- **Action:** 1. Pushed down the side clips on the RAM slots to pull out the memory sticks. 2. Unscrewed the GPU from the expansion slot and pressed the PCIe release latch to remove it safely. 3. Unscrewed the SATA SSD/HDD from the drive cage and slid it out.

Step 5: Removing the Motherboard

- **Action:** Finally, I removed the mounting screws holding the motherboard to the cabinet standoffs and gently lifted the entire board out, placing it on an anti-static surface.

Task 3:

PC Reassembly Process and Installation Order

Below is the step-by-step documented process of reassembling the PC, along with an explanation of why the specific installation order matters:

Step 1: Installing the Motherboard into the Cabinet

- **Action:** I carefully placed the motherboard into the empty cabinet, aligned it with the I/O shield at the back, and screwed it securely into the standoffs.
- **Why this order?** The motherboard is the central hub. It must be installed first because all other components connect directly to it. If we put the power supply or drives first, they might block our hands from screwing the motherboard properly.

Step 2: Installing the RAM and Storage Drives

- **Action:** I inserted the RAM sticks into the DIMM slots until they clicked. Then, I slid the SATA SSD into the drive cage and secured it with screws.

- **Why this order?** It is much easier to push the RAM into the slots while there is empty space. Doing this before installing large components like the GPU prevents our hands from hitting other parts.

Step 3: Installing the Graphics Card (GPU)

- **Action:** I aligned the GPU with the main PCIe x16 slot, pressed it in until the latch locked, and screwed the metal bracket to the cabinet.
- **Why this order?** The GPU is usually the bulkiest component. Installing it late ensures it doesn't block access to the small SATA ports or the lower half of the motherboard during the earlier steps.

Step 4: Connecting the Internal Power and Data Cables

- **Action:** I connected the main 24-pin ATX power cable, the 8-pin CPU power cable, and the SATA data cables from the storage drive to the motherboard.
- **Why this order?** Wiring should always be done after the components are physically secured. This allows for proper cable management and ensures no wires

get trapped under the hardware.

Step 5: Closing the Panel and External Connections

- **Action:** Finally, I slid the side panel back on, secured the thumb screws, and reconnected the mouse, keyboard, monitor, and the main AC power cord to test the boot.

Task 4:

Safety Precautions and Tools Checklist for PC Assembly/Disassembly

1. Essential Tools Checklist:

- **Phillips Head Screwdriver (Magnetic tip preferred):** Needed to remove and secure cabinet screws, motherboard standoffs, and drives. A magnetic tip is important so that tiny screws do not fall and get lost inside the case.
- **Anti-Static Wrist Strap or Mat:** Used to ground the technician's body. This is important to prevent Electrostatic Discharge (ESD) from frying sensitive microchips.
- **Compressed Air Can / Anti-static Blower:** Required to safely clear out built-up dust from the chassis, fans, and heat sinks before reassembly without physically touching the delicate fins.

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- **Thermal Paste and 99% Isopropyl Alcohol:**
Essential for cleaning off old, dried thermal compound and applying a fresh layer on the CPU to ensure optimal thermal conductivity and cooling.
- **Tweezers or Small Pliers:** Helpful for picking up dropped screws from tight, hard-to-reach corners of the motherboard cabinet.

- **2. Safety Precautions Checklist:**
- **Complete Power Isolation:** Always turn off the SMPS switch and completely unplug the main AC power cord from the wall outlet. Why? To ensure there is zero risk of electric shock and to prevent short-circuiting the motherboard components.
- **Discharge Static Electricity:** Always touch an unpainted metal part of the PC case or wear an anti-static wrist strap before touching any internal part. Why? Human bodies carry static electricity that can instantly destroy sensitive components like RAM or the CPU processor.
- **Handle Components by the Edges:** Never touch the golden connector pins on the bottom of the CPU, RAM sticks, or Graphics Card. Why? Natural oils from our skin can cause corrosion or create poor electrical connectivity.
- **Avoid Working on Carpets:** Always work on a clean, hard, wooden or plastic table. Why? Carpets generate a massive amount of static electricity through friction, which is highly dangerous for exposed hardware.
- **Never Force Components:** If a RAM stick, GPU, or power cable isn't sliding into its slot easily, do not

use excessive force. Why? Applying too much pressure can permanently bend the slot pins or snap the motherboard PCB.

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